Summary in Graph

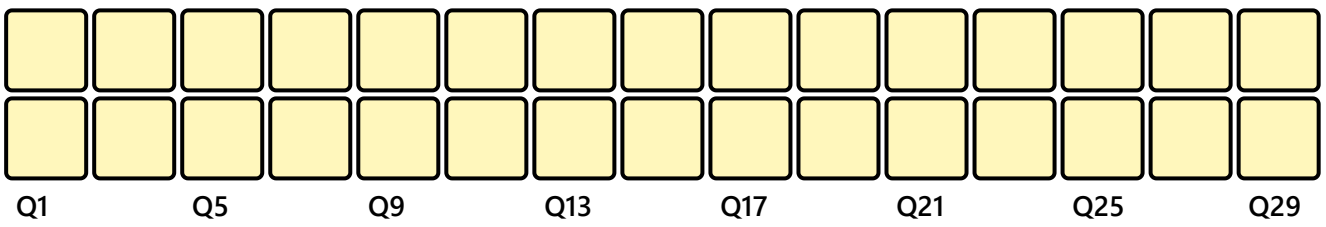
Exam Summary (GATE Overflow CS Test Series 2026 | Operating Systems | Test 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (7) out of 65 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Operating System

Which of the following CPU scheduling algorithms suffer from convoy effect? (Mark all the appropriate choices)

A. First Come First Served
B. Non-preemptive Shortest Remaining Time First
C. Round Robin Scheduling

D. Preemptive Shortest Remaining Time First

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

The degree of multi-programming in an operating system is affected by
(Mark all the appropriate choices)

A. long-term scheduler
B. short-term scheduler
C. medium-term scheduler
D. job scheduler

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. Kernel-level threads are handled by the operating system directly
B. Kernel level threads are faster to create and manage than user level threads
C. Multiple threads of the same process can be scheduled on different processors in kernel-level threads.
D. If a kernel-level thread is blocked, all threads of the same process also gets blocked

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following statements about CPU scheduling algorithms is/are TRUE? (Mark all the appropriate choices)

A. Shortest Job First (SJF) algorithm is optimal with respect to the average turn around time.
B. SJF algorithm is optimal with respect to the average waiting time.
C. SJF algorithm may lead to starvation.
D. SJF algorithm guarantees a better or same average turn around time, as First Come First Served algorithm

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct about deadlocks in an Operating System? (Mark all the appropriate choices)

A. If a system is in a safe state, it will never enter a deadlock after a sequence of resource allocations.
B. If a system is in an unsafe state, it will eventually enter a deadlock.
C. Deadlock prevention is stricter that deadlock avoidance.
D. Deadlock prevention ensures better resource utilization than deadlock avoidance.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

The banker's algorithm is being run in a system with m resource classes and n processes. In the limit of large m and n, the number of operations that must be performed to check a state for safety is proportional to $n^a(\lg n)^b m^c(\lg m)^d$. The value of $2a+3b+5c+7d=$_____

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

A counting semaphore is initialized to X. Then 13 P (wait) operations and 25 V (signal) operations are completed on this semaphore. If the final value of semaphore is 8, then the value X will be _____

Your Answer: | Correct Answer: -4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. If each resource type has exactly one instance, then a cycle in resource allocation graph is a necessary condition for the existence of deadlock
B. If each resource type has exactly one instance, then a cycle in the resource allocation graph is a sufficient condition for the existence of deadlock
C. If each resource type has several instances, then a cycle in the resource allocation graph is a sufficient condition for the existence of deadlock
D. If each resource type has several instances, then a cycle in the resource allocation graph is a necessary condition for the existence of deadlock

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Processes in an operating system can be in any of the following states. Choose the proper matching for the given table.

	Process State		**Description**
$a.$	Ready	$i.$	The process is waiting for some event to occur (such as an I/O completion or reception of signal).
$b.$	Waiting	$ii.$	The process is waiting to be assigned to a processor.
$c.$	Terminated	$iii.$	Instructions are being executed.
$d.$	Running	$iv.$	The process has finished execution.
		$v.$	The process is being created.

A. $a-ii, b-i, c-iv, d-iii$
B. $a-v, b-i, c-iv, d-iii$
C. $a-ii, b-i, c-iv, d-v$
D. $a-iii, b-i, c-iv, d-ii$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. Switching the CPU to another process requires performing a state save of the current process and a state restore of a different process.
B. Whenever an interrupt arrives, the CPU kills the currently executing process.
C. A typical Operating System allows a process to create threads but not another process.
D. Only the processes in Ready state are stored in the job queue.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. The Data section is reserved for local variables when they are declared.
B. The Heap is made up of the global and static variables, allocated and initialized prior to executing the main.
C. The space on the Stack is used for the dynamic memory allocation, and is managed via calls to new, delete, malloc, free, etc.
D. The Text section is made up of the compiled program code, read in from non-volatile storage when the program is launched.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A process executes the following segment of code :

```
#include <stdio.h>
#include <unistd.h>
int main()
{
    fork();
    fork() && (fork() || fork());
    printf("GO 2020");
}
```

The number of times "GO 2020" will be printed is _____

Your Answer: | Correct Answer: 8 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A process executes the following segment of code :

```
if(fork() && fork() || fork()){
    printf("GATE2021");
}
```

The number of times "GATE2021" will be printed is _____

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following code segment:

```
pid_t pid;
pid = fork();
if (pid == 0) {
thread create(...);
fork();
}
fork();
```

If a is the total number of processes created, and b is the total number of threads created when the above code is executed, the value of $a + b =$ _____

Your Answer: | Correct Answer: 13 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A process executes the following segment of code :

```
int foo(int n)
{
    int i;
    for(i=0; i <n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(j < i)
            {
                fork();
            }
        }
    }
    return 0;
}
```

The total number of child processes created if foo is called with parameter $n = 5$ is _____

Your Answer: | Correct Answer: 1023 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following synchronization construct used by two the processes

```
turn1 = true;
turn2 = true;
Process 1
...
while(true)
{
    turn1 = true;
    while(turn2);
    /* Critical Section */
    turn1 = false;
}
...
Process 2
...
while(true)
{
    turn2 = true;
    while(turn1);
    /* Critical Section */
    turn2 = false;
}
...
```

Which one of the following statements is/are TRUE?

A. The proposed solution prevents deadlock but fails to guarantee mutual exclusion.
B. The proposed solution guarantees mutual exclusion but fails to prevent deadlock.
C. The proposed solution guarantees mutual exclusion and prevents deadlock.
D. The proposed solution fails to prevent deadlock and fails to guarantee mutual exclusion.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following two-process synchronization solution.

```
P0:
while(TRUE) {
  while(turn!=0);
  critical_section();
  turn = 1;
  noncritical_section();
}
P1:
while(TRUE) {
  while(turn!=1);
  critical_section();
  turn = 0;
  noncritical_section();
}
```

The shared variable turn is initialized to zero. Which one of the following is TRUE?

A. This is a correct two-process synchronization solution satisfying mutual-exclusion, progress and bounded waiting.
B. This solution violates mutual exclusion requirement.
C. This solution violates progress requirement.
D. This solution violates bounded wait requirement.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following synchronization solution.

```
void swap (bool *a, bool *b)
{
    bool temp = *a;
    *a = *b;
    *b = temp:
}
do {
    key = TRUE;
    while ( key == TRUE) swap (&lock, &key);
    <Critical Section>
    lock = FALSE;
} while (TRUE);
```

If the boolean shared variable lock is initialized to FALSE and all the assignment statements happen atomically in hardware, which of the following statements is/are correct?

A. Race condition may be possible.
B. Mutual exclusion is guaranteed.
C. Progress is guaranteed.
D. Starvation freedom is guaranteed.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following two process synchronization solution. Assume all assignments are atomic.

```
flag[0] = flag[1] = FALSE;
P0:
do {
    flag[0] = TRUE;
    turn = 1;
    while (flag[1] && (turn == 1));
    critical section
    flag[0] = FALSE;
    remainder section
} while (TRUE);

P1:
do {
    flag[1] = TRUE;
    turn = 0;
    while (flag[0] && (turn == 0));
    critical section
    flag[1] = FALSE;
    remainder section
} while (TRUE);
```

Which of the following is/are correct? (Mark all the appropriate choices)

A. Race condition may be possible.
B. Mutual exclusion is guaranteed.
C. Progress is guaranteed.

D. Bounded waiting is guaranteed.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following multi-process synchronization solution.

```
do {
while (test_set(&lock));

lock = FALSE;

} while (TRUE);
```

Assume that the shared boolean variable lock is initialized to FALSE and the test_set function is an atomic function which returns the current value of the argument and sets it to TRUE. Which of the following is/are correct? (Mark all the appropriate choices)

A. Race condition may be possible.
B. Mutual exclusion is guaranteed.
C. Progress is guaranteed.
D. Bounded waiting is guaranteed.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

The following program consists of 3 concurrent processes and 3 binary semaphores. The semaphores are initialized as $S_0 = 0, S_1 = 1$ and $S_2 = 1$.

Process P_0	Process P_1	Process P_2

while (true) {
wait (S_0);
print '1';
release (S_1);
release (S_2);
}
	wait (S_1);
	release (S_0);
		wait (S_2);
		release (S_0);

How many times will process P_0 print '1'?

A. At least twice
B. Exactly once
C. Exactly twice
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following process synchronization program involving N processes.

```
semaphore S = 21;
Process 1
S.signal();
while (1) {
  // do something
  S.wait();
  //critical section
  S.signal();
  // do something
}
Process 2..N
while (1) {
  // do something
  S.wait();
  //critical section
  S.signal();
  // do something
}
```

The maximum number of processes that can be in the critical section at any moment of time is _____

A. 21
B. 22
C. 23
D. More than 23

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following code using semaphores.

```
semaphore S1 = 1, S2 = 1;
int cnt1 = cnt2 = 2;
Process 1
while (cnt1--) {
  // do something
  S1.wait();
  print{"GATE"};
  S2.signal();
  // do something
}

Process 2
while (cnt2--) {
  // do something
  S2.wait();
  print{"Overflow"};
  S1.signal();
  // do something
}
```

The output of the above program will be _____ (Mark all the appropriate choices)

A. GATE Overflow GATE Overflow
B. Overflow GATE Overflow GATE
C. GATE GATE Overflow Overflow
D. Overflow GATE GATE Overflow

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 01sec | Discuss

Q #24 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following Producer-Consumer implementation using a buffer of size N. Assume that initially the buffer is empty.

```
Semaphore mutex = __;
Semaphore empty = __;
Semaphore full = __;
```

```
producer {
  while (1) {
    Produce new resource;
    wait(empty);
    wait(mutex);
    Add resource to an empty buffer;
    signal(mutex);
    signal(full);
  }
}
consumer {
  while (1) {
    wait(full);
    wait(mutex);
    Remove resource from a full buffer;
    signal(mutex);
    signal(empty);
    Consume resource;
  }
}
```

For the correct working of the above code the initialization of the semaphores mutex, empty and full must be as

A. $\text{mutex} = 0, \text{empty} = \text{N}, \text{full} = 1$
B. $\text{mutex} = 1, \text{empty} = 1, \text{full} = \text{N}$
C. $\text{mutex} = 1, \text{empty} = \text{N}, \text{full} = 0$
D. $\text{mutex} = 0, \text{empty} = \text{N}, \text{full} = \text{N}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider the following solution to the classical Dining Philosophers Problem using Semaphores, where 5 philosophers in a dining table are trying to eat using 5 chopsticks each one placed to the right and left of each philosopher.
semaphores chopstick $[0 \ldots 4]$ all initialized to 1.

```
repeat
wait(chopstick[i+(i%2)]);
wait(chopstick[(i+1-(i%2)) mod 5]);
. . .
eat
. . .
signal(chopstick[i]);
signal(chopstick[(i+1) mod 5]);
. . .
think
. . .
until false;
```

Which of the following properties are satisfied by the above solution? (Mark all the appropriate choices)

A. Starvation freedom
B. Deadlock freedom
C. Mutual Exclusion
D. Progress

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider a system having k resources of the same type. These resources are shared by 5 processes $A, B, C, D,$ and E which have peak demands of $5, 7, 9, 3$ and 8 respectively. Further it is restricted that at any point of time processes A and B together cannot hold more than 8 resources whereas no such restrictions are there for processes C, D and E. For which of the following values of k, deadlock may occur? (Mark all the appropriate choices)

A. 24
B. 25
C. 27
D. 26

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider the following snapshot of a system running n processes.

Process i is holding x_i instances of a resource $R, 1 \leq i \leq n$. Currently, all instances of R are occupied. Further, for all i, process i has placed a request for an additional y_i instances while holding the x_i instances it already has. There are exactly two processes p and q and such that $y_p = y_q = 0$. Which one of the following can serve as a sufficient condition to guarantee that the system would not be going to a deadlock?

A. $x_p + x_q > \max_{k \neq p,q} y_k$
B. $\max(x_p, x_q) > 1$
C. $x_p + x_q \geq \min_{k \neq p,q} y_k$
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

5 processes P_0 through P_4; 3 resource types A (10 instances), B (5 instances), and C (7 instances)
Snapshot at time T_0 :

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P_0	0	1	0	7	5	3	3	3	2
P_1	2	0	0	3	2	2			
P_2	3	0	2	9	0	2			
P_3	2	1	1	2	2	2			
P_4	0	0	2	4	3	3			

The system is currently in a safe state. Consider the following independent requests for additional resources in the current state:

REQ1: P_0 requests 0 units of A, 2 units of B and 0 units of C
REQ2: P_1 requests 1 units of A, 0 units of B and 2 units of C
REQ3: P_4 requests 3 units of A, 3 units of B and 0 units of C
Which one of the following is TRUE?

A. Only REQ1 and REQ2 can be permitted.
B. Only REQ2 and REQ3 can be permitted.
C. All of $\text{REQ1, REQ2 and REQ3}$ can be permitted.
D. None of $\text{REQ1, REQ2 or REQ3}$ can be permitted.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

A system has four processes and five allocatable resources. The current allocation and maximum needs are as follows:

	Allocation					Maximum					Available				
	A	B	C	D	E	A	B	C	D	E	A	B	C	D	E
P_0	1	0	2	1	1	1	1	2	1	3	0	0	x	1	0
P_1	2	0	1	1	0	2	2	2	1	1					
P_2	1	1	0	1	0	2	1	3	1	0					
P_3	1	1	1	1	0	1	1	2	2	1					

The smallest value of x for which the given state is a safe state, is _____

A. 1
B. 2
C. 3
D. None of the above

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following set of processes, with the arrival times and the CPU-burst times gives in milliseconds.

Process	Arrival Time	Burst Time
P_1	0	8
P_2	1	4
P_3	3	3
P_4	4	2

The average turnaround time for these processes with the preemptive shortest remaining time first (SRTF) algorithm is _____

Your Answer: | Correct Answer: 7.75 | Not Attempted | Time taken: 00min 00sec | Discuss

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